

The original Borda count and partial voting

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Abstract In a Borda count, BC, M. de Borda suggested the last preference cast should receive 1 point, the voter's penultimate ranking should get 2 points, and so on. Today, however, points are often awarded to (first, second,..., last) preferences cast as per $(n, n-1, \dots, 1)$ or more frequently, $(n-1, n-2, \dots, 0)$. If partial voting¹ is allowed, and if a first preference is to be given n or $n-1$ points regardless of how many preferences the voter casts, he/she will be incentivised to rank only one option/candidate. If everyone acts in this way, the BC metamorphoses into a plurality vote... which de Borda criticized at length. If all the voters submit full ballots, the outcome—social choice or ranking—will be the same under any of the above three counting procedures. In the event of one or more persons submitting a partial vote, however, outcomes may vary considerably. This preliminary paper suggests research should consider partial voting. The author examines the consequences of the various rules so far advocated and then purports that M. de Borda, in using his formula, was perhaps more astute than the science has hitherto recognised.

Abbreviations

AV (= IRV = STV)	alternative vote
IRV (= AV = STV)	instant run-off voting
PR	proportional representation
STV (= AV = IRV)	single transferable vote
BC	Borda count
MBC	modified BC
QBS	quota Borda system
TRS	two-round voting

¹ In a ballot on n options/candidates, a partial vote is one in which the voter ranks only m options/candidates, where $1 \leq m < n$.

1 Background

In decision-making, numerous forums use (simple, weighted, twin, qualified or consociational) majority votes. The question posed is thus invariably a dichotomy, so those concerned—members of the given electorate or legislature—may either vote in favour, or vote against, or abstain; in these instances, there can be no partial voting. Some countries use multi-option voting in referendums—New Zealand, for example, held a five-option ballot in 1992—but most of these polls have been conducted in a form of two-round voting, TRS, so here too, in any one round, there cannot be any partial voting.

Some electoral systems, however—approval voting, AV (alternative vote), BC and QBS, some PR-list systems and PR-STV—allow the voter a degree of pluralism. The PR-list system used in Belgium allows the voter to cast more than one preference, but only for candidates of one party. In the version of PR-list used in Luxembourg and Switzerland, the electorate in any one constituency may support as many candidates as there are to be representatives therein, and such votes may be cross-party. The alternative vote, AV, as in Australia and Papua New Guinea, is preferential, as is PR-STV, the chosen system for Malta and both jurisdictions in Ireland: these two varieties of STV also allow the voter to support more than one candidate of more than one party.

Aware of the dangers of ethnic voting, Papua New Guinea's form of AV lays down a minimum number of preferences but not a maximum—to be valid, a vote must contain at least three preferences. Even where voting is compulsory, as in Australia, full ballots are not obligatory. In Irish practice, North and South, few vote for only one candidate; many vote for only one party; a considerable number rank several candidates but few if any submit full ballots; indeed, in a six-seater constituency, for example, there would be little purpose in so doing.² Partial voting, therefore, is part of the real world. Accordingly, *inter alia*, research work on the Borda count, BC, should also consider partial voting.

2 The story so far

Although McLean and Urken (1995) suggest that maybe Ramon Lull was the first to think about a BC, there is no doubt about the work of Nicholas of Cusa. In 1435, on the subject of electing the Holy Roman Emperor, Cusanus proposed the voter “puts after each name... a number ranging between one and the total number of candidates. Thus if there are ten candidates, the best would have the number ten after his name, and the worst would be number 1... The candidate who gets the highest number of [points]... is elected.” In effect, then, the cardinal suggested $(n, n - 1, \dots, 1)$. In a

² Conscious of the fact that a voter may regard the difference between his/her first and second ranking more passionately than the difference between his/her fifth and sixth, say, Nauru uses a unique counting rule: $(1, 1/2, \dots, 1/n)$.

Table 1 Profile from *Mémoire sur les élections au scrutin*

Preferences	No. of voters			
	1	7	7	6
First preference	a	a	b	c
Second preference	b	c	c	b
Third preference	c	b	a	a

personal note he adds, “With much study, I have not been able to find a safer system, and believe me, no more perfect system can be found,”³ (Sigmund 1963).

It seems that his proposal contained no provision for any partial votes. He did, however, add one small proviso: “each of the electors gets a ballot with the name of each candidate upon it (except his own, if he is running),” (*ibid.*). There may well have been different ballot papers, some of nine candidates and some of ten, (or maybe some of ten and others of eleven), but those voting were expected to rank all the candidates, or so it would seem.

Jean Charles de Borda advocated a different procedure: starting with the last preference, he argued, points should be awarded according to the rule: a , $a + b$, $a + 2b$...etc.; he further proposed that a should equal 1 and b should also equal 1. Thus, in an n option/candidate ballot, the voter’s last ranking gets 1 point, his/her penultimate gets 2 points, and so on, up to the first preference which (in a full ballot) gets n points; this translates into $(1, 2, \dots, n)$, Cusanus’s rule back to front.

Today, however, a commonly accepted procedure reverts, not to Cusanus’s original $(n, n - 1, \dots, 1)$ but, more usually, to a variation thereof: $(n - 1, n - 2, \dots, 0)$. I have not (yet) identified the person who first devised the latter, but the earliest reference found so far is in Black (1958).

If everyone has cast full ballots, there is no difference to the outcome in any of these counting procedures. If any of the voters have submitted partial votes, however, the difference can be profound.

3 de Borda and partial voting

In his *Mémoire sur les élections au scrutin*, (1781), M. de Borda considered the above profile, here shown in a somewhat easier format, with all 21 voters casting full ballots on the three candidates, a , b and c (Table 1).

He makes no mention of partial voting. Arrow (1963) suggests this is deliberate and that de Borda’s “major point... is that the entire ordering of the individual voters among alternative candidates is needed.” In contrast, Saari (2008), argues that the voter who “votes for only two candidates gives them, respectively, 2, 1 points”, but he too doubts “whether Borda wrote anything about a partial count.”⁴

³ Other voting systems known to exist at this time include (i) majority voting, which dates from Greek or even Sumerian times; (ii) plurality voting, advocated by Pliny the Younger in AD 105; (iii) a forerunner of the Condorcet procedure proposed by Lull; and (iv) a form of approval voting, first used in Venice in 1268.

⁴ Private correspondence.

Table 2 Interpretations of a partial vote: first preference c .

Methodology	Points awarded					
	a	b	c	d	e	f
BC untreated ($n, n-1, \dots, 1$)	0	0	6	0	0	0
BC untreated ($n-1, n-2, \dots, 0$)	0	0	5	0	0	0
BC averaged	3	3	6	3	3	3
	2	2	5	2	2	2
MBC	0	0	1	0	0	0

4 The effects of partial voting

There are several ways in which a partial vote might be counted, including one in which Michael [Dummett \(1997\)](#) proposes that the other options/candidates should each get an average share of the points which the voter has not exercised; a second proposal, the Modified BC, MBC,⁵ (of which, more in a moment), in effect stipulates de Borda's rule; [Emerson \(1994\)](#).

Consider, then, a vote on six options/candidates— a, b, c, d, e and f . If a voter casts only one preference for, let us say, option/candidate c , points awarded could be as shown in Table 2.

In a 'BC untreated', the voter will be incentivised to cast only a first preference, for thus his favourite option/candidate would gain a 6 or 5 point advantage over all the other options/candidates. When a 'BC averaged' is used, the voter has a smaller 3-point advantage, so is still incentivised to truncate her vote. The MBC, however, is neutral, for the voter's x th preference, if cast, always gets 1 point more than an $(x+1)$ th preference, regardless of whether or not he/she has cast that $(x+1)$ th preference. ([Emerson 2007](#)). The overall effect of the MBC, then, is to encourage participation and to the full,⁶ for he who in an MBC ranks only one option/candidate gains a 1-point advantage over all the other options/candidates, whereas she who ranks all the options/candidates gains a 1-point advantage over her second preference, a 2-point advantage over her third choice, and so on.

de Borda (1781), was highly critical of the plurality vote. It would seem that either he wanted the voter to cast a full ballot, and this is the more likely conclusion: that or, if partial voting were to be allowed, it should be done under the above-mentioned rule identified by [Saari \(2008\)](#). A third possibility is that he did not think that some

⁵ In effect, the MBC formula is the same as that noted in ([Saari 2008](#), p. 197).

⁶ This has often been seen in practice. The author's first demonstration of this methodology was in Belfast in 1986 when over 200 persons—members of nearly every political party, from *Sinn Féin* to the Ulster Unionists—took part in an 'experiment in consensus'. To get such a broad range of participation, a full eight years before the cease-fire, was achievement enough. But to find their consensus was an even greater success: "Northern Ireland to have devolution and power-sharing under a tripartite Belfast–Dublin–London agreement"—it was, as it were, a mini-Belfast Agreement, 12 years ahead of its time. The author has used the MBC on many other occasions since then, not only in Ireland and the UK, but also in the USA, in what was the USSR, and in mainland Europe, not least in Bosnia.

people would submit partial votes. When the BC was put into practice in the Academy, however, “the voters [soon] found how to manipulate the Borda rule... by truncating their lists” (McLean and Urken 1995). When this and other deficits of his voting procedure were pointed out, de Borda responded by saying “My election method is only for honest men,” (*ibid.*); this suggests the third most probable explanation: that he did not consider the possibility of partial voting.

5 The MBC

In a vote on n options/candidates, voters may rank (one, some or) all n options/candidates. If $n > 10$, say, then a number of preferences less than 10 should be stipulated as the number required to be cast for a full ballot.⁷

Accordingly, the rule is as follows. In an MBC in which voters are asked to rank up to n options/candidates, points should be awarded to (first, second, ..., last) preferences as follows:

$$(m, m - 1, \dots, 1), \quad (1)$$

where m is the number of preferences cast:

$$1 \leq m \leq n \quad (2)$$

This rule thus caters for both full and partial votes; it was devised by the de Borda Institute, mainly as a result of experience gained in Northern Ireland and the Balkans where, because of passions held and sufferings endured, procedures *must* allow voters to submit partial ballots.

But maybe the MBC is what de Borda actually had in mind when he suggested $a, a + b, a + 2b$, and so on. Granted, the person(s) who proposed $(n, n - 1, \dots, 1)$ and $(n - 1, n - 2, \dots, 0)$ may be forgiven for what might initially appear to be an obvious simplification. As explained above, however, this interpretation only holds true if and when all the voters submit full ballots.

Now just as many people oppose compulsory voting, so too many would argue that, not least in conflict zones, voters must be allowed (to abstain, or) to submit either a partial or a full vote, as they wish. Such is the case in an MBC: people may submit a full ballot, or a partial ballot, or of course they may abstain. To deal with these three categories in reverse order, those who abstain have no influence on the outcome; those who vote partially have a partial influence, and on a progressive scale—he who ranks one option/candidate exercises 1 point, she who ranks two exercises 3 points, and so on; while those who submit a full ballot exercise the sum of the series $\{1, \dots, n\}$ points and thus have a full influence.

Granted, there are countless hypothetical instances in which a set of voters' partial rankings will produce a different outcome to that which comes from an analysis of

⁷ In any process of identifying a social ranking, when voters are prioritising, say, the top six options from a choice of twenty, they may be asked to rank, let us say, just six options.

their full rankings. It is submitted, however, that a full ranking more accurately reflects the voter's opinion. Furthermore, in practice, experience suggests that an MBC does indeed incentivise the voter to submit a full, or at least a fairly full ballot, and a preferential voting procedure is obviously more difficult to manipulate than any single preference system. This practice, which now extends over 30 years in both 'normal' societies and in some of the world's conflict zones, is commensurate with "research [which] makes it arguable that only the Borda count offers an accurate accounting of the voters' preferences." (Saari 2001).

6 Conclusion

Democracy, as it were by definition, is for everybody. The democratic process should not be an adversarial procedure in which a supposed majority wins everything and a minority wins nothing. Rather, in decision-making, it should be a process by which all can influence that which then becomes the confluence; an inclusive process to identify at least the best possible compromise, and at most, the collective wisdom. To do this, the voting procedure should allow if not encourage those concerned to participate, and to the full; in other words, it should allow for partial voting. Scientific research should therefore study this possibility, and it may yet emerge that, in suggesting his formula, M. de Borda was wiser than has so far been acknowledged.

Acknowledgements The BC was originally proposed, if not by Ramon Lull circa 1300, then by Nicholas Cusanus in the fifteenth century. The first to give this methodology a scientific analysis, however, was of course Jean-Charles de Borda in 1781, but others, often independently, have also devised this methodology: Hother Hage in 1860 or thereabouts, Rev. Dodgson in 1884, and the present author in 1978, to name but three. Variations of this methodology have been or are still being used in Kiribati, Nauru, and Slovenia for elections; in Britain and Ireland by a few companies, NGOs and at least one political party—the Irish Green Party—again, mainly for elections; occasionally in Porto Allegre, Brazil, for prioritisations in participatory budgeting; and quite frequently in non-political fields, Euro-vision song contests, *Grand Prix* motor racing, and so on.

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